

# Earth-State Embeddings: self-supervised representations of the observed ocean, probed against the Atlantic overturning circulation

blauewelt/earth — technical report, v6

16 August 2026

## Abstract

We train a small self-supervised model to compress everything the observing system knows about each point of the ocean, each month, into a single vector — an *embedding* — and then ask whether those vectors, which never see any transport measurement, carry the Atlantic Meridional Overturning Circulation (AMOC). They do: a linear read-out from the embeddings along 26.5°N attains a year-blocked cross-validated correlation of  $r = 0.60$  with the monthly, deseasonalized RAPID transport series — to our knowledge the first cross-validated monthly-scale figure against real RAPID data from inputs that never include RAPID. A temporal model over the frozen embeddings predicts the ocean’s anomaly field beyond a year ahead (anomaly correlation 0.62 at one month, crossing the conventional 0.5 useful-skill line near four months), while the transport read-out loses skill after about one season, consistent with its wind-driven monthly variance. This revision adds a *probe ladder* — ridge, small MLP, and a supervised attention head over the *unpooled* section — and it rewrites the story of the week. Pointwise nonlinearity recovers nothing anywhere: the MLP trails the ridge on every target, for every codec. But the attention head, which sees the  $\sim 67$  section pixels individually instead of their mean, raises every codec — and the codec whose channel tokens carry  $3 \times 3$  spatial patches reaches  $r = \mathbf{0.672}$  across two independently trained seeds (0.690 and 0.654), RMSE  $\approx 2.1$  Sv against a 2.79 Sv target, the best transport figure of the programme. The mechanism is thermal wind: geostrophic transport is the east-minus-west density difference across the section, and mean-pooling computes exactly the statistic in which that difference cancels. An end-to-end control then decomposes the headline: the same head over *raw* section data reaches 0.659 at matched receptive field, so pretraining’s own margin on this task is +0.013 — indistinguishable from zero, and a directional +0.031 reported from the first seed alone is withdrawn here — the codec’s demonstrated value lies in field prediction and transfer, not in the monthly RAPID probe. An apparent finding that expanding the input from 14 to 24–25 channels had *cost* monthly RAPID skill is retracted as a representation claim — the pooled probes were billing relocated information as lost — while the genuine gain those channels brought at the deep MOVE array at 16°N ( $r = 0.206$  [0.044, 0.346], our first multi-target confidence interval excluding zero) stands.

This revision (v6) reports the programme’s move to the *rolled forecast at scale* (§8), and its central finding is about capacity. The temporal model now reads a *stencil* of distant neighbour embeddings alongside the pixel’s own history; skill is scored both one step ahead (forecast ratio against persistence) and rolled twelve months over the AMOC corridor (corridor AUC), with a frozen gate head that must reproduce its pinned score before any evaluation counts. At the original stage-2 size ( $576 \times 8$ , 34M parameters) the two scores *anti-correlated* across input geometries — the best one-step arms rolled worst — and a systematic audit cleared the instrument and localized the effect in the model-under-iteration. The resolution is capacity: at  $768 \times 12$  (88M) the same wide-input geometry that lost at 34M gains +0.050 corridor AUC paired by seed, and at  $1024 \times 16$  (205M) the corridor AUC reaches  $\mathbf{0.664/0.663/0.664}$  across three seeds — against 0.589 for the no-neighbour baseline and 0.604 for the best small-model geometry — with no sign of saturation. On the one-step forecast, parameters beat every other axis measured: input width (Fibonacci sunflower layouts,  $34 \rightarrow 89$  points,  $\approx -0.002$  per rung, then *saturating* at 89 — 144 points buy nothing) and step budget ( $60k \rightarrow 200k$ , decelerating toward an asymptote) both pay, and both pay less. **On the rolled forecast, scale is not the largest effect.** Adding Gaussian noise to the model’s own inputs at its measured one-step error scale is worth +0.057 corridor AUC at fixed capacity — enough that a noised 88M model beats the clean 205M one — and a near-field 8-point stencil is worth +0.035 over wide-and-far inputs at equal parameters, reversing the forecast-axis ranking. A geometric measurement plausibly explains both: at 4444 km reach a head’s dependency cone covers the entire model domain by horizon three and about half its input slots fall outside it, so every rolled number here rests on holding the world beyond the window at climatology. The small-

model geometry ranking that occupied a week of experiments is preserved in Appendix A as a capacity-starvation artifact.

## 1 Introduction

The AMOC moves on the order of 17 Sv of water northward through the Atlantic’s upper kilometre ( $1 \text{ Sv} = 10^6 \text{ m}^3 \text{ s}^{-1}$ ), carries roughly a petawatt of heat, and is directly measured at only a handful of latitudes, for only about two decades. It is the canonical example of a climate-critical quantity that is *composite*: part wind-driven (Ekman transport), part density-driven (geostrophic shear), part boundary currents. No single satellite or float sees it; it must be inferred from the state of the ocean around it.

This project asks a representation-learning question of that inference problem: *if we compress the whole observable ocean state into generic, task-blind embeddings, do climate-critical integrals like the AMOC come along for free?* The design is deliberately strict: the embedding model never sees any transport data; every read-out is a linear (ridge) regression, so any skill must already be present, linearly accessible, in the representation; and every claim is cross-validated in blocks that respect the autocorrelation of the climate system. The project also serves as a live test of a foundation-model hypothesis (§6): that a single representation trained on *all* the data — every ocean, every channel — serves specific regional tasks at least as well as one trained only on the task’s own region.

## 2 Terminology

Terms are used throughout in the following precise senses.

**Transport** the physical quantity all “truth” targets measure: the volume of water a current system moves, in Sverdrups ( $1 \text{ Sv} = 10^6 \text{ m}^3 \text{ s}^{-1}$ ). The AMOC’s strength is a transport ( $\sim 17 \text{ Sv}$  northward across  $26.5^\circ \text{N}$ ). A *transport series* is one array’s monthly record of it; none is ever a model input.

**Fold** one round of the year-blocked cross-validation: a calendar year serves as the test set while the probe is re-fitted from scratch on all other years. “Per fold” means exactly that re-fitting — no test month ever influences the weights that predict it.

**Transport claims / field-prediction claims** the project’s two scoreboards, kept apart because their baselines differ. Transport claims: predicting the mooring series (k-fold  $r$ , RMSE in Sv, event capture) against the wind-only ridge. Field-prediction claims: predicting the ocean’s own input fields ahead in time (chan%, rollout MSSS/ACC) against persistence and climatology.

**Channel** one physical input field: e.g. surface current speed, mixed-layer depth, temperature at 700 dbar, zonal wind stress. The data tensor has 12–15 channels in the first generation, 24–25 in the second, and 39 in the third (Table 1, §3.2).

**Tensor generation** one build of the input tensor whose channel set, grid, or time axis differs from the others enough that results are not comparable across it (§3.2). Three exist, and we name them by what defines them: the **coarse tensor** ( $1^\circ$ , 4 Argo pressure levels; “family 1” in earlier notes), the **deep tensor** ( $1^\circ$ , 8 Argo levels; “family 2”), and the **quarter-degree tensor** ( $0.25^\circ$  North Atlantic, 16 Argo levels, sea-surface height added, time axis extended to 1982; “family 3”). Run names keep the short prefixes (**f3\_** etc.); prose uses the descriptive names.

**Pixel** one ocean grid cell —  $1^\circ \times 1^\circ$  in the first two tensor generations,  $0.25^\circ \times 0.25^\circ$  in the third. A *pixel-month* is one pixel at one month — the atomic sample.

**Channel space** the space of the physical fields themselves. An error “in channel space” is an error in predicted physical values (normalized units), directly comparable with a persistence forecast of the same fields.

**Embedding / embedding space** the 64-dimensional vector  $z$  the encoder produces for a pixel-month, and the space of such vectors. An error “in embedding space” compares predicted to actual  $z$ ; it is an internal diagnostic, since  $z$  has no physical units.

**Codec** the stage-1 encoder–decoder (§4); it encodes a pixel-month’s channels to  $z$  and can decode  $z$  back to channels. “Codec” emphasizes that it is a learned compression.

**Anomaly space** all dynamic channels are expressed as departures from that pixel’s own monthly climatology (computed over training years only), then normalized. This removes everything predictable from location and calendar alone.

**Climatology** the long-term mean for a given pixel and calendar month. As a forecast, “climatology” predicts zero anomaly — the no-skill floor every real forecast must beat.

**Persistence** the forecast that the anomaly stays exactly as it is now. Hard to beat at short range; decays into a poor forecast at long range. *Damped persistence* multiplies the anomaly by its own lag- $h$  autocorrelation — the fair statistical baseline.

**Patch codec** a codec variant whose channel token carries the  $3 \times 3$  neighbourhood of the pixel (nine values and nine observed-flags) rather than a single scalar, giving the encoder gradient vision and turning the representation from an expander into a genuine compressor (§4).

**Stage 2 / temporal model** a causal transformer over a pixel’s sequence of frozen embeddings that predicts next month’s embedding. “Frozen” means the codec’s weights are not updated by stage-2 training, so improvements are attributable.

**Probe** a read-out fitted from frozen embeddings to a target series; the codec and stage 2 are never updated by it. Historically a ridge regression, deliberately weak; now the bottom rung of the probe ladder below. A probe measures what a representation *contains*.

**Probe ladder** three read-outs of the same frozen embeddings, increasing only in what they may see: *ridge* (linear, on the section-mean embedding), *MLP* (adds pointwise nonlinearity, same pooled input), *attention head* (adds spatial structure: one learned query attends over the unpooled per-pixel embeddings of the section). Each rung localizes where a capability comes from.

**Section** the row of grid cells nearest a mooring array’s latitude, clipped to the array’s longitude span (e.g. RAPID: 26.5°N, 80°W–13°W). Probes read the mean embedding along the target’s section.

**Holdout years** entire calendar years (2009, 2017, 2023) excluded from all training; with a held-out mid-Atlantic longitude band they form the blocked validation splits. Random splits are never used — adjacent months are too correlated to count as independent.

**k-fold (year-blocked)** every calendar year takes one turn as the test set while the probe is fit on the others; the assembled out-of-fold predictions are scored once. Confidence intervals come from a bootstrap that resamples whole years (a *block bootstrap*).

**chan%** held-out next-month prediction error reduction versus persistence, in channel space, in anomaly space:  $100(1 - \text{MSE}_{\text{model}}/\text{MSE}_{\text{pers}})$ . Comparable only within one channel set, since the persistence baseline moves when channels change.

**MSSS** mean squared skill score,  $1 - \text{MSE}/\text{MSE}_{\text{ref}}$  [6]; our rollout reports it against climatology, persistence, and damped persistence.

**ACC** anomaly correlation coefficient: the Pearson correlation between predicted and observed anomalies at a given forecast horizon.  $\text{ACC} = 0.5$  is the conventional “useful skill” line.

**RAPID / FC / MOVE / OSNAP / SAMBA** the five transport truth series: the RAPID array at 26.5°N (2004–, monthly), the Florida Current cable at 27°N (1982–, daily), MOVE at 16°N (2000–2022, the *deep* western-boundary flow), OSNAP in the subpolar gyre (2014–2022), SAMBA at 34.5°S (2009–2017). None is ever an input.

**Stencil** the set of *neighbour pixels* whose embedding sequences the temporal model reads alongside the centre pixel’s own (§8). A stencil of  $S$  slots turns the model’s input from one 24-month embedding sequence into  $S+1$  of them. Geometries used: rings at a fixed radius, spirals (every point on its own bearing), and *sunflower* layouts — golden-angle spirals with a far-heavy radial ramp, at Fibonacci counts 34/55/89/144, reaching 111–4444 km. A neighbour that falls on land or off-window is a *dead slot*, encoded as zero.

**Forecast ratio** the one-step scoreboard of the stencil programme:  $\text{MSE}_{\text{model}}/\text{MSE}_{\text{pers}}$  for  $z_{t+1}$  prediction on held-out windows, in embedding space. Lower is better; persistence sits at 1.0 by construction and the measured persistence MSE is  $\approx 3.1$   $z$ -units<sup>2</sup>. Reproduces to  $\pm 0.002$  across seeds, so paired differences of 0.002 are readable.

| channel(s)                                     | source               | record | role                 |
|------------------------------------------------|----------------------|--------|----------------------|
| surface current speed, MLD                     | GLORYS reanalysis    | 1993–  | dynamics             |
| sea-surface height ( $\frac{1}{4}^\circ$ only) | GLORYS reanalysis    | 1993–  | geostrophy           |
| T at 4 / 8 / 16 pressure levels                | Roemmich–Gilson Argo | 2004–  | density structure    |
| S at 4 / 8 / 16 pressure levels                | Roemmich–Gilson Argo | 2004–  | density structure    |
| wind stress $\tau_x, \tau_y$                   | NCEP R1              | 1948–  | Ekman forcing        |
| within-month $\tau$ std. dev.                  | NCEP R1 daily        | 1948–  | storminess           |
| monthly SST (optional, $1^\circ$ only)         | OISST v2.1           | 1981–  | thermal memory       |
| SST, precip climatologies ( $1^\circ$ only)    | OISST, GPCP          | fixed  | (measured inert; §9) |

Table 1: Channels of the data tensor across its three generations (§3.2). Coarse and deep tensors: monthly,  $1^\circ$ , 1993–present; NA pilot window ( $100^\circ\text{W}$ – $20^\circ\text{E}$ ,  $0$ – $70^\circ\text{N}$ ; 5,787 ocean pixels) or global (44,964 pixels). Quarter-degree tensor: monthly,  $0.25^\circ$ , NA window, 84,405 ocean pixels, axis extended to 1982–01 (channels absent before their records enter as missing tokens),  $C=39$ . Missing data (Argo before 2004, polar gaps, pre-1993 months) is represented explicitly, never imputed.

**Corridor** the scored subset of the rolled evaluation: the fastest quarter of the window’s pixels by train-month mean current speed, dilated two cells, plus the RAPID section — 29,627 of 84,405 window pixels. It is where the AMOC’s limbs live, and it is rendered as a globe layer in the companion visualization.

**Corridor AUC** the rolled scoreboard: initialize with true embeddings, roll twelve months feeding predictions back, decode, and score MSSS against climatology over the corridor’s pixel-months at each horizon,

$$\text{AUC} = \frac{1}{12} \sum_{h=1}^{12} \left( 1 - \frac{\text{MSE}_h}{\text{MSE}_{\text{clim},h}} \right).$$

“AUC” is *area under the skill-versus-horizon curve* — a plain mean over twelve monthly horizons — **not** a ROC statistic. Higher is better; climatology sits at 0 by construction, and the ranking is unchanged under persistence and damped-persistence references and under horizon re-weighting (§8.2). Table 4 is the reference table.

**Gate head** a frozen reference head (e017, no neighbours) re-evaluated at the start of every corridor evaluation; it must reproduce its pinned AUC of 0.643 to the third decimal or the evaluation is void. Every number in §8 passed this gate.

**Capacity tiers** the three stage-2 sizes of §8, with parameters measured from the checkpoints: *base*  $576 \times 8$  (34.1M at 56 slots), *big*  $768 \times 12$  (88.0M), *xl*  $1024 \times 16$  (205.4M).

### 3 Data

The tensor (Table 1) is built entirely from credential-free public archives. Two properties matter most. First, *missingness is information*: a pixel-month’s unobserved channels enter the encoder as learned “missing” tokens rather than imputed values, so the representation knows what the observing system knew. Second, everything is in anomaly space (§2) — the founding negative result of the project was that a model trained on raw fields wins reconstruction by memorizing the seasonal cycle while learning nothing about change.

The five truth series (RAPID, FC, MOVE, OSNAP, SAMBA) ride along as never-input targets: 240 RAPID months plus 792 additional truth months across four other latitudes.

#### 3.1 The sources, one by one

Every input is a credential-free public archive; this subsection says what each one is, what channels it contributes, and why it is in the tensor. (The companion visualization at [blauewelt.github.io/earth](https://blauewelt.github.io/earth) renders most of these sources as interactive globe layers with per-dataset documentation; the descriptions here follow its catalog.)

**GLORYS ocean reanalysis (CMEMS).** An eddy-permitting model of the global ocean constrained by satellite altimetry, SST, sea ice, and in situ profiles — the closest thing oceanography has to a best-estimate movie of the ocean since 1993. Contributes monthly *surface current speed* ( $\sqrt{u^2 + v^2}$ ; the western boundary currents that carry the AMOC’s upper limb are current-speed structures) and *mixed-layer depth* (log-transformed; deep winter mixing in the subpolar gyre is where surface waters acquire the density to sink — the “engine room” of the overturning). At  $1^\circ$  these come from the quarter-degree ensemble member binned down; the quarter-degree tensor reads the native grid and adds *sea-surface height*, whose meridional gradient is the geostrophic surface flow and whose boundary difference is the classical altimetry AMOC proxy [1].

**Roemmich–Gilson Argo climatology.** The gridded monthly temperature and salinity fields ( $1^\circ$ , 0–2000 dbar) built from the Argo float array [21] — the backbone subsurface observing system:  $\sim 4,000$  autonomous floats, each profiling every  $\sim 10$  days. Contributes T and S at 4, 8, or 16 pressure levels (10–1900 dbar) depending on tensor generation. These are the density channels: thermal wind makes the transport an east-minus-west density difference, and the deep levels see the cold limb the surface channels cannot. Intrinsically  $1^\circ$ -smooth (in the quarter-degree tensor they are bilinearly upsampled and carry no sub-degree information — stated, not hidden). The record begins in 2004; earlier months are missing tokens.

**NCEP/NCAR Reanalysis 1 surface wind stress.** The longest-running atmospheric reanalysis [22], daily since 1948, on a Gaussian grid regridded to the tensor. Contributes monthly-mean  $\tau_x, \tau_y$  (Ekman transport, the wind-driven part of the AMOC’s monthly variability — and hence also the source of the wind-only baseline every codec must beat) and the *within-month standard deviation* of the daily stress (storminess: buoyancy loss in storms drives convection in a way monthly means erase). Its length is what lets the quarter-degree tensor’s axis start in 1982.

**NOAA OISST v2.1.** Daily  $0.25^\circ$  SST from AVHRR plus in situ data since late 1981 [23], monthly-meaned; optional channel 15 of the deep tensor, and (as a fixed 1991–2020 climatology) one of the two deliberately inert channels of §9.

**GPCP v2.3 precipitation climatology.** The satellite–gauge merged precipitation record [24]; enters only as the second inert climatology channel of §9.

**Truth arrays (never inputs).** Five basin-scale transport series, each a moored or cable observing system: *RAPID–MOCHA*  $26.5^\circ\text{N}$  [25], the flagship full-depth AMOC record (2004–, our 240 labelled months); the *Florida Current submarine-cable transport* [26], the lowest-latency AMOC component, daily since 1982 — the decade 1982–92 is usable truth only for the quarter-degree tensor’s extended axis; *MOVE*  $16^\circ\text{N}$  [27], the longest deep-limb record (southward NADW flow, 2000–); *OSNAP* [28], the subpolar array where overturning variability originates (2014–); and *SAMBA*  $34.5^\circ\text{S}$  [29], the only sustained South Atlantic array (2009–, anomalies only).

### 3.2 Tensor generations, and why they may not be compared

The tensor is rebuilt from the fetcher scripts at the start of every training job, so a change to a fetcher silently changes the input of every subsequent run. Sampling 8 Argo pressure levels instead of 4 took the tensor from 14 channels to 24 (25 with monthly SST) without any dispatch asking for it. Every result below therefore belongs to one of three *tensor generations* — the **coarse tensor** ( $1^\circ$ , 4 Argo levels), the **deep tensor** ( $1^\circ$ , 8 levels), and the **quarter-degree tensor** ( $0.25^\circ$  NA, 16 levels, SSH, axis from 1982) — and cross-generation comparison is not licensed: the persistence baseline that `chan%` is measured against moves with the channel set, the wind-only ridge baseline moves with the grid (0.531 on the  $1^\circ$  tensors, 0.568 on the quarter-degree), and, as §7.5 shows, so does the meaning of the transport probe. (Earlier project notes call these “channel families” 1–3; run names keep the shorter prefixes.) Channel counts in this report are read from each checkpoint, never from a run’s name — a run directory carried a misleading name for an hour before this was enforced, and every run now ships a provenance manifest recording the tensor it actually saw.

## 4 Methods

**Stage 1 — the codec.** A per-pixel masked autoencoder producing  $z \in \mathbb{R}^{64}$  per pixel-month, trained on all pixel-months outside the holdout blocks. The pilot is 0.92M parameters; a 10.25M variant ( $320 \times 8$ ,

the size the Chinchilla-style arithmetic of the Compute paragraph anchors for the current tensor) is training at time of writing. Checkpoints carry their architecture, and every loader reconstructs from it.

**What exactly enters the encoder.** The encoder is a small transformer ( $128 \times 4$  in the pilot,  $320 \times 8$  in the 10M variant) over  $C+2$  tokens for one pixel-month:

- **One token per channel.** If the channel is observed and not masked, the token is a linear projection of its scalar value (in anomaly space: the departure from that pixel’s own monthly climatology, standardized) plus a learned per-channel identity embedding. If the channel is *masked for training*, the value is replaced by a learned **mask** token; if it is *genuinely unobserved* — Argo before 2004, polar gaps — by a distinct learned **missing** token. “I am hiding this from you” and “the observing system never measured this” are different facts, and the distinction measurably matters.
- **One context token:** a linear projection of  $[\sin(2\pi m/12), \cos(2\pi m/12), \text{lat}/90, \text{lon}/180]$  — season and position. Never masked. This token is why static climatology input channels carry zero information (§9).
- **One CLS token**, whose final hidden state is projected to  $z \in \mathbb{R}^{64}$ .

The decoder never sees the inputs — it answers *queries* (channel,  $\Delta\text{lon}$ ,  $\Delta\text{lat}$ ,  $\Delta\text{month}$ ) from  $z$  alone, with offsets up to  $\pm 3$  cells and months, so the training signal is reconstruction of masked channels *plus* prediction of spatial and temporal neighbours.

**The patch codec.** At 14 channels a pixel-month carries on average 10.4 observed values, so a 64-dimensional  $z$  is an *expander* ( $\sim 6\times$ ), as in masked language models; the information bottleneck is the masking objective, not the dimensionality. The patch variant gives each channel token its  $3 \times 3$  neighbourhood — nine values and nine observed-flags, projected by  $\text{Linear}(2p^2, d_{\text{model}})$  — so that at 24 channels roughly 160 observed values enter one 64-dimensional vector, a genuine 2.5:1 compression. Longitude wraps (the globe is periodic in  $x$ ) and latitude clamps with out-of-range rows marked unobserved; the centre cell defines whether a channel counts as observed, and reconstruction targets remain centre values, so the loss is unchanged and  $p=1$  checkpoints load identically. The motivation is physical as well as informational: thermal wind *is* a density gradient, and a gradient does not exist within a single pixel.

**Stage 2 — the temporal model.** A causal transformer (default  $d_{\text{model}}=96 \times 3$  layers, 0.35M parameters; capacity sweep in Fig. 9) over each pixel’s 24-month embedding sequence, predicting  $z_{t+1}$ ; scored after decoding to channel space. The codec stays frozen.

**Probes — the ladder.** The base instrument is a ridge regression from section-mean embeddings to each truth series, deseasonalized, regularization chosen on a training tail — never on test data; verification is year-blocked k-fold with block-bootstrap confidence intervals. Two stronger read-outs sit above it, behind the same folds and the same inner-tail discipline. A small MLP ( $64 \rightarrow 32 \rightarrow 1$ , weight decay, early stopping, three seeds per fold) tests whether skill is present but not *linearly* accessible. An attention head tests whether skill is present but not accessible from the *pooled* section: every (pixel, month) embedding enters as a token with a longitude encoding, one learned query cross-attends over them, and a two-layer read-out maps the pooled vector to transport ( $\sim 25\text{k}$  parameters, weight decay  $10^{-2}$ , token dropout, three seeds per fold). The codec stays frozen throughout — these are probes, not fine-tuning. The design principle: each rung adds one capability, so a gap between rungs is attributable to that capability alone. Beneath the ladder sit two end-to-end *controls*: the identical head fed raw section values and observed-flags instead of embeddings, at single-pixel and at  $3 \times 3$  receptive fields — pretraining is then the only manipulated variable (§7.3).

**Rollout.** For horizons  $h = 1 \dots 12$ : initialize with true embeddings, predict one month, feed the prediction back, repeat — scored in channel space against climatology, persistence, and damped persistence, and probed for the AMOC at each horizon.

**Compute.** Runs through 2026-08-07 were trained on GitHub-hosted 4-core CPU runners at  $\sim 1.3$  steps/s. Since then three rented consumer GPUs (RTX 4090,  $\sim \$0.24/\text{h}$  each) are attached as self-hosted runners at  $\sim 124$  steps/s each — a factor of 95 per box, which turns a 30k-step schedule from six hours into four minutes and made the controlled comparisons of §7 affordable. The raw source data ( $\sim 13$  GB) is mirrored as a GitHub release so a fresh box seeds its cache from a CDN rather than the origin archives, one of which (the Argo host) times out under load. Nothing about the model or protocol changed with the hardware. Chinchilla-style anchor for sizing: the current tensor carries  $\sim 270\text{M}$  observed values per epoch; at the  $\sim 20:1$  data-to-params ratio that suggests a  $\sim 13\text{M}$ -parameter codec —  $14\times$  the pilot, and the reason the 10M variant exists. The arithmetic is redone whenever the data changes.

## 5 Metrics and their statistical power

Full definitions are in the glossary; the reliability ordering is the substance. **chan%** is decision-grade within a channel set: seed spread  $\pm 0.5$  points, so differences  $\geq 2$  points are real. The **fixed-holdout probe**  $r$  (36 held-out months, effective sample  $\approx 15$  after autocorrelation) has a 95% CI of  $\pm 0.57$  — a compass, never a verdict; the same configuration has drawn 0.09 and 0.33 on two seeds. The **year-blocked k-fold** tests all 240 RAPID months once each ( $\text{SE} \approx 0.10$ ) and is the only probe number we argue from. The most instructive failure: a  $d_z=128$  codec set the all-time fixed-holdout record (0.528) while the k-fold read 0.166 — the coarse instrument flatters widest exactly where overfitting is easiest.

The stage-2 programme of §8 adds two instruments at opposite ends of the power spectrum. The **forecast ratio** is the sharpest number in the project:

$$\text{ratio} = \frac{\text{MSE}(\hat{z}_{t+1}, z_{t+1})}{\text{MSE}(z_t, z_{t+1})}$$

over held-out windows — the temporal model’s one-step error in embedding space, divided by the error of simply asserting that next month’s embedding equals this month’s (persistence, whose measured MSE is  $\approx 3.1$   $z$ -units<sup>2</sup>). A ratio of 0.13 therefore reads “the model’s one-step error is 13% of the no-change forecast’s.” Because every stage-2 arm shares the same frozen codec, the denominator is identical across arms and the ratio is directly comparable across the whole of §8 — which is exactly why it, and not a channel-space score, is the ladder’s screening metric. Its statistical power is what makes the paired designs work: three-seed spread is  $\pm 0.002$ , so paired same-seed differences of 0.002 (the width effect) are readable and differences of 0.03 (the capacity effect) are  $\sim 30\sigma$ . Its limitation is inherited from its space: it is an internal, embedding-space quantity, blind by construction to anything the frozen codec cannot represent, and a gain on it need not survive iteration — which is what the **corridor AUC** exists to test. That instrument is the opposite trade: physically meaningful (channel-space MSSS against climatology, rolled twelve months, averaged over horizons, scored on the 29,627 corridor pixels), expensive (hours per head), and guarded rather than powerful — its error discipline is the gate head’s exact reproduction (0.643, third decimal, every run) plus paired seeds, and its decision threshold is the observed seed spread ( $\pm 0.005$  at the 88M tier,  $\pm 0.001$  at 205M). Every claim in §8 names which of the two instruments it is read from.

## 6 Related work

Three literatures border this project; we take them in turn. (The repository file `m1/LITERATURE.md` carries the full survey and is updated more often than this section.)

**AMOC reconstruction from observables.** The classical line regresses the RAPID transport on physically chosen observables. Frajka-Williams [1] reconstructs the MOC from satellite sea-level anomaly plus the cable and Ekman terms and explains  $>90\%$  of variance ( $r \approx 0.95$ ) — but after 18-month low-pass filtering, with in-sample calibration; Sanchez-Franks et al. [2] reach  $r = 0.83$  from altimetry via a thermal-wind argument under the same filtering; Worthington et al. [8] fit boundary densities to the upper-mid-ocean transport ( $R^2 = 0.78$  monthly, held-out  $r \approx 0.75$  on a single 2017–18 test year). Fingerprint methods [9] recover multi-decadal AMOC trends from SST patterns but no monthly signal. The machine-learning line trains almost exclusively on simulations: Solodoch et al. [3] report  $r \approx 0.98$

| study                                                                                  | inputs                               | target & filtering                 | protocol                         | skill                           |
|----------------------------------------------------------------------------------------|--------------------------------------|------------------------------------|----------------------------------|---------------------------------|
| Frajka-Williams [1]                                                                    | altimetry + <b>cable</b> + Ekman     | RAPID MOC, 18-mo low-pass          | in-sample                        | $r \approx 0.95$                |
| Sanchez-Franks [2]                                                                     | altimetry (thermal wind)             | RAPID, deseas. + 18-mo             | in-sample                        | $r = 0.83$                      |
| Worthington [8]                                                                        | boundary densities                   | RAPID <i>UMO</i> , monthly         | 1 test year                      | $R^2 = 0.78$ ; $r \approx 0.75$ |
| Caesar [9]                                                                             | subpolar SST fingerprint             | AMOC index, 20-yr LOWESS           | model-calibrated                 | trend only                      |
| Solodoch [3]                                                                           | OBP+SSH+SST+ $\tau$                  | <b>ECCO</b> MOC, monthly           | chrono. 70/30, <i>simulation</i> | $r \approx 0.98$                |
| Meng [4]                                                                               | SSH+OBP maps                         | eddying-model MOC                  | <i>simulation</i>                | skill $\approx 0.44$            |
| Wölker [5]                                                                             | virtual Argo + $\tau$ + cable        | VIKING20X 26.5°N                   | <i>simulation</i>                | $R^2 \approx 0.74$              |
| <b>this work</b> (ridge)                                                               | 14-ch embedding, <b>no transport</b> | <b>real</b> RAPID, monthly deseas. | year-blocked $k$ -fold           | $r = 0.602$ [0.46, 0.73]        |
| <b>this work</b> (head)                                                                | 24-ch embedding, <b>no transport</b> | <b>real</b> RAPID, monthly deseas. | year-blocked $k$ -fold, 2 seeds  | $r = 0.672$                     |
| <i>same, under their filter</i>                                                        | —                                    | +18-mo low-pass                    | <i>as above</i>                  | $r = 0.33$ –0.75                |
| <i>classical inputs re-scored in OUR protocol (§6.2) — same 227 months, same folds</i> |                                      |                                    |                                  |                                 |
| wind only                                                                              | section $\tau_x, \tau_y$             | real RAPID, monthly                | year-blocked $k$ -fold           | $r = 0.401$ [0.27, 0.56]        |
| cable only                                                                             | <b>Florida Current transport</b>     | real RAPID, monthly                | year-blocked $k$ -fold           | $r = 0.254$ [0.15, 0.36]        |
| cable + wind                                                                           | <b>two of RAPID's three terms</b>    | real RAPID, monthly                | year-blocked $k$ -fold           | $r = 0.566$ [0.45, 0.68]        |

Table 2: **Joint results table.** Every row is a defensible number in its own frame; the frames differ so much that the rightmost column should not be read as a ranking. §6.1 explains each difference and quantifies the largest. Bold marks the two design choices that most change the task: whether a *transport measurement* is among the inputs, and whether the target is the *real* ocean or a simulation.

for a CNN reading OBP/SSH/SST/wind strips — inside the laminar 1° ECCO state estimate; in eddy-resolving models the same task drops to explained variance 0.44–0.74 [4, 5], which we read as the honest difficulty scale. Application to real RAPID data at monthly cadence with blocked cross-validation is, as far as we can determine, absent from the literature. Our pooled-ridge  $r = 0.60$  — monthly, deseasonalized, real data, year-blocked  $k$ -fold, no RAPID-derived inputs — sits where the eddying simulations predict it should (the attention-head read-out reaches 0.65–0.69 across two seeds, with the attribution caveats of §7.3), and no published surface-to-OSNAP reconstruction exists for our OSNAP null to contradict.

## 6.1 Why these numbers are not comparable

Table 2 spans  $r = 0.44$  to  $r = 0.98$ , and almost none of that spread is model quality. Four mechanisms, in descending order of how much they matter.

**1. Some inputs already contain the answer.** RAPID does not measure one thing; it sums three: the Florida Current transport, the wind-driven Ekman transport, and the upper-mid-ocean geostrophic transport. Frajka-Williams [1] takes the *cable* record (which *is* the Florida Current term, measured) and Ekman (from wind) as inputs, and reconstructs only the third. Two of the three addends are handed to the model; the reported correlation is substantially the correlation of a partial sum with its own total. Wölker [5] likewise includes cable transport. Our inputs contain no transport measurement of any kind — not the cable, not Ekman as a transport — so we are inferring all three terms from the ocean state. These are different problems, and the harder one should be expected to score lower.

**2. The 18-month filter removes 83% of the variance and 87% of the degrees of freedom.** This is the difference most often glossed as a methodological detail, so we measured it on the RAPID series itself. Monthly deseasonalized RAPID has  $\sigma = 2.79$  Sv, a lag-1 autocorrelation of 0.32, and an integral timescale of 3.5 months — so its 240 months carry  $\approx 68$  effective degrees of freedom. After an 18-month low-pass, the integral timescale becomes 25 months and the effective sample size falls to  $\approx 9$ , while the retained variance is **17%** of the original. Two consequences. First, the filtered task is easier: what survives is the slow, heat-content-like component that altimetry and density observables track well, while the wind-driven month-to-month variance — the part we are scored on — has been removed. Second, the filtered statistic is barely constrained:  $r = 0.95$  on 9 effective points carries a 95% confidence interval of roughly  $[0.77, 0.99]$ , and  $r = 0.83$  one of  $[0.37, 0.96]$  — intervals wide enough to overlap our unfiltered number. (As a check on the arithmetic: the same calculation gives  $[0.42, 0.73]$  for our  $r = 0.60$  at 68 effective points, against the  $[0.46, 0.73]$  our block bootstrap reports independently.)

We can also show the instability empirically rather than asserting it. Two codecs of *identical* configuration differing only in training seed score 0.543 and 0.531 monthly — a gap of 0.012 — but 0.449 and 0.327 once 18-month filtered, a gap of 0.122, ten times larger. Across our runs the filtered correlation ranges from 0.33 to 0.75 while the monthly one stays within 0.53–0.62. We report the filtered number for every run (it is in the repository’s leaderboard), but we decline to headline a statistic whose seed noise is an order of magnitude larger than the effect anyone wants to measure.

**3. In-sample calibration is a fit, not a prediction.** The classical reconstructions [1, 2] tune their coefficients on the same period over which skill is then reported; Caesar [9] calibrates against models. No held-out set exists, so the number answers “how well can this functional form describe these years” rather than “what would it say about a year it has not seen”. Worthington [8] is the honourable exception in the classical line, holding out 2017–18 — one test year, where our protocol re-fits from scratch for every year in turn. Our year-blocked  $k$ -fold is the strictest arrangement here and costs us skill relative to every in-sample row, by construction.

**4. A simulated ocean is a different ocean.** Solodoch’s  $r \approx 0.98$  [3] is obtained inside ECCO, a  $1^\circ$  state estimate whose dynamics are laminar and whose surface fields and MOC are generated by the same model — the mapping to be learned is close to deterministic and free of observational noise. The moment the same task is posed in eddy-resolving simulations it falls to 0.44–0.74 [4, 5], and the real ocean adds sparse, irregular sampling on top of eddies. We read the eddying-model range, not the laminar one, as the relevant difficulty scale — and our 0.60–0.67 on real observations sits inside it.

## 6.2 Making them comparable: the classical inputs under our protocol

A like-for-like comparison requires real RAPID data, monthly resolution without low-pass filtering, no transport measurement among the inputs, and an out-of-sample protocol; no published result meets all four. But the comparison can be rescued from the other end. Rather than dress our number up in the literature’s conventions, we put the literature’s *inputs* through our protocol — same target, same deseasonalization, same year-blocked folds, same block bootstrap — so that the rows differ only in what the model is allowed to see. The Florida Current cable, normally one of our never-input truth series, is used as an input for these rows and these rows only (`ml/classical.baseline.py`).

The bottom block of Table 2 reports it, on the 227 months where RAPID and the cable overlap. Wind stress alone gives  $r = 0.401$ ; the cable alone  $r = 0.254$ ; **the two together — i.e. two of RAPID’s three constituent terms, handed to the model as direct measurements — give  $r = 0.566$   $[0.45, 0.68]$ .** Our embedding, which is shown no transport measurement of any kind, reaches 0.578–0.620 under the same pooled ridge and 0.672 with the unpooled head.

We read that carefully. It does *not* say we beat Frajka-Williams: that reconstruction also uses altimetry for the third term, which this ladder omits, and it targets a filtered series. What it does say is sharper and is the number we could not get any other way: at monthly resolution, out of sample, being handed the Florida Current and the Ekman transport outright is worth about  $r = 0.57$  — and a task-blind embedding of the ocean state matches it without being handed anything. The gap between that 0.57 and a published 0.95 is therefore not model quality; it is the 18-month filter and the in-sample fit, quantified. A supporting observation in the same direction: when we apply the 18-month filter to these classical rows’ *out-of-fold* predictions their correlations collapse ( $-0.07, -0.23, +0.24$ ), which is

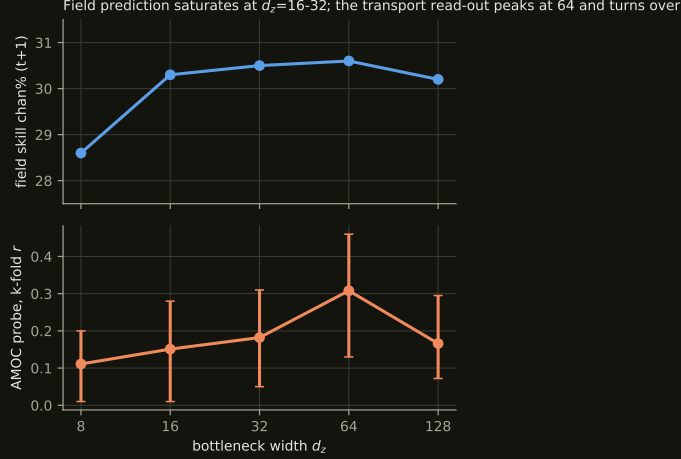


Figure 1: Bottleneck width  $d_z$  versus its two customers, on the 12-channel North-Atlantic tensor at matched training steps. Field prediction (top) saturates by  $d_z=16-32$ ; the AMOC probe (bottom, with 95% block-bootstrap CIs) rises to a peak at  $d_z=64$  and turns over at 128 — 128 ridge features against  $\sim 220$  truth months per fold overfits.  $d_z=64$  became the working setting, a choice made when the tensor had 12 channels.

what a filtered statistic with nine effective points does once it is no longer allowed to see its own test years.

**Ocean and climate forecasting.** Our field-prediction metrics are the standard decadal-verification quantities under other names: skill vs. climatology is MSSS with a climatological reference, and we adopt ACC per lead and damped persistence from the same framework [6]. Reference points at our monthly cadence: Ham et al.’s ENSO CNN [10] holds Niño3.4 correlation  $> 0.5$  to  $\sim 17$  months (the ocean’s most predictable index); Taylor & Feng [11] forecast global monthly SST anomaly fields with a Unet-LSTM (RMSE  $0.48^\circ\text{C}$  at 1 month rising to  $0.63$  at 18) — the closest published analogue to our stage-2 rollouts. The daily  $1/4^\circ-1/12^\circ$  neural ocean forecasters (XiHe, WenHai, GLONET) and the global weather models they descend from [12, 13] operate in a different regime (days, full fields, reanalysis-supervised) and are cited for orientation, not comparison.

**Self-supervised and foundation models for Earth observation.** Architecturally we are a masked autoencoder [14] over channel tokens rather than image patches. The flagship Earth-system foundation models — ClimaX [15], Aurora [16], and the land-surface line of Prithvi [17] and SatMAE [18] — evaluate mainly by full fine-tuning; frozen-representation evaluation, our entire protocol, is the exception: Prithvi WxC’s anomaly-space masked pretraining with frozen backbone and small heads is our closest architectural relative (atmosphere-only), and Lehmann et al. [7] argue explicitly that frozen-backbone extension should be a standard foundation-model quality metric. AlphaEarth Foundations [19] embeds every terrestrial 10 m pixel into 64 dimensions — the same width we use — from ten data modalities, and is the clearest statement of the “embedding field” idea this project transplants to the ocean; it is terrestrial-only, and its modality list (translated to ocean: altimetry, gravimetry, scatterometry, ocean colour, sea ice) is our data-ladder shopping list. Model sizing follows the compute-optimal-scaling heuristic of Hoffmann et al. [20] adapted to observed-value counts (§4, Compute). To our knowledge no self-supervised foundation model exists for the gridded ocean interior; frozen-embedding probes to integrated circulation indices, and missingness-as-information over the historical observing system, remain unoccupied territory.



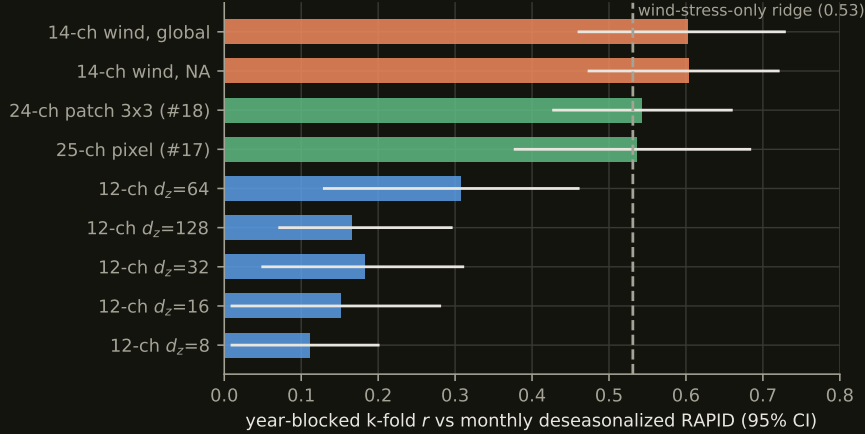


Figure 2: Year-blocked k-fold correlation with monthly deseasonalized RAPID transport, all codec generations, 95% CIs. Adding wind stress (orange) nearly doubled the defensible probe; training globally cost the Atlantic read-out nothing. The deep-tensor codecs (green) fall back onto the dashed wind-only baseline — a ridge on section-mean wind stress with no embedding at all.

## 7 Results

### 7.1 The bottleneck: what width buys

Fig. 1: the transport read-out wants more width than field reconstruction rewards, but not unboundedly — the turnover at  $d_z=128$  is a truth-data limit, not a representation limit. That this sweep was run at  $C=12$  becomes important in §7.5.

### 7.2 The probe ranking, and what the wind already knows

Fig. 2 is the project’s headline and its conscience in one picture. The wind channels moved the probe more than every architecture decision combined ( $0.31 \rightarrow 0.60$ ); the global codec matches the regional one exactly ( $0.602$  vs  $0.604$  — the generic-embedding hypothesis holds at fixed capacity); and the wind-only baseline at  $0.53$  says most monthly skill is Ekman physics. The coarse tensor’s embedding increment over pure wind is  $+0.07$ ; its distinctive contributions are the low-frequency component (18-month low-passed  $r = 0.64$ ) and event anatomy (Fig. 6). In Sverdrups: RMSE  $2.23$  Sv against a target standard deviation of  $2.79$  Sv.

### 7.3 The probe ladder: pooling was hiding the signal

Fig. 3 is this revision’s central result, and it changes both a number and a story. The number: the patch codec read by the attention head reaches  $r = 0.690$  [ $0.570, 0.781$ ], RMSE  $2.02$  Sv (and  $0.654$  on a second codec seed, below) — against  $0.60$  for the best pooled-linear configuration, and against a wind-only baseline of  $0.531$  that the pooled deep-tensor probes had barely cleared. The story: the ladder’s two rungs dissociate cleanly. Pointwise nonlinearity (MLP rung) recovers nothing on any target for any codec — so the ridge was never the limit as a *function class*. Spatial structure (head rung) recovers a great deal — so the limit was the *pooling*. Mean-pooling the section computes the one statistic in which an east-west difference cancels, and thermal wind says the transport *is* an east-west difference. The two architectural changes of this cycle turn out to be halves of one physical idea: the  $3 \times 3$  patch puts local gradients into each embedding, the attention head reads gradients across the section, and each looked worthless without the other — the patch codec’s pooled numbers were indistinguishable from its pixel control.

**The attribution matrix.** An end-to-end baseline completes the picture: the *same* attention head, same folds, same regularization, fed raw section anomalies (values + observed-flags) instead of embeddings, at both receptive fields:

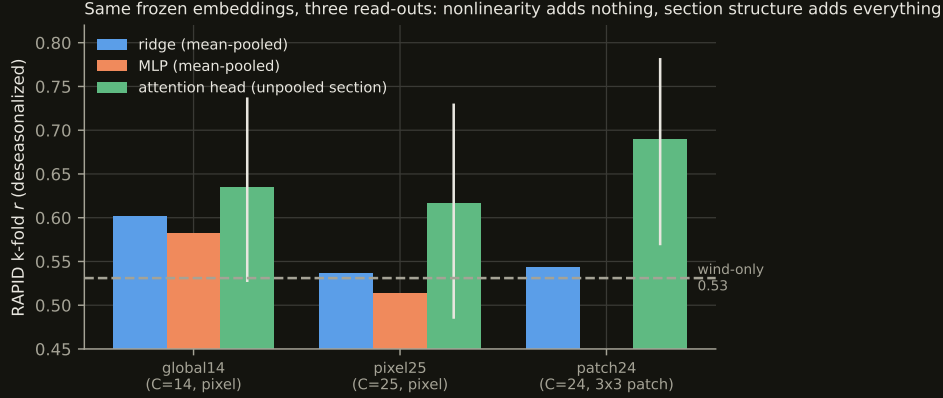


Figure 3: Same frozen embeddings, three read-outs, same year-blocked folds. The MLP (orange) trails the ridge (blue) everywhere — pointwise nonlinearity recovers nothing. The attention head over the unpooled section (green, with 95% CIs) beats every pooled probe, restores the 25-channel codec to parity with the 14-channel one, and reveals the patch codec — invisible to pooled probes — as the strongest representation in the programme at  $r = 0.690$ .

| tokens carry               | raw data           | codec embedding         |
|----------------------------|--------------------|-------------------------|
| single pixel               | 0.613 [0.48, 0.73] | 0.617 [0.49, 0.73]      |
| $3 \times 3$ neighbourhood | 0.659 [0.55, 0.75] | 0.690 / 0.654 (2 seeds) |

**A second seed, and a retraction.** The 0.690 cell was replicated on an independently trained codec of identical configuration. It came back at 0.654 [0.543, 0.748]. The capability claim survives — both seeds beat every pooled probe and the wind-only baseline decisively, so unpooling is real — but the *pretraining margin does not*. The raw- $3 \times 3$  baseline is codec-independent at 0.659; the two-seed embedding mean is 0.672; the advantage is therefore +0.013, which is nothing. We previously reported +0.031 “positive in direction, not significant” and we withdraw even that directional reading: it was one seed’s noise. Note that the pooled ridge replicated far more tightly across the same two codecs (0.543 vs 0.531) than the head did (0.690 vs 0.654) — the more expressive probe is also the noisier instrument, and no future head number in this programme should be quoted from a single seed.

What remains, then, decomposes as: *unpooling* the section is the whole of the improvement (pooled ridge 0.54  $\rightarrow$  head 0.65+); the  $3 \times 3$  *receptive field* adds  $\sim +0.045$  and does so on raw data alone; *pretraining* adds nothing measurable at either receptive field. For the monthly RAPID probe, a supervised head over raw section data recovers everything the codec offers. The pretraining case now rests entirely on the scoreboards where 240 labelled months have no answer: field prediction (stage 2 beats persistence everywhere; the matching end-to-end field-forecaster baseline is the open follow-up), transfer to sparsely-labelled targets like MOVE, and whether a margin appears at all once the codec is sized to its Chinchilla anchor.

That last test has now run, on the quarter-degree tensor at 40.7M parameters (§7.4), and it neither rescues the pretraining case nor closes it: the head reads 0.662 [0.557, 0.745] from the codec against 0.628 [0.514, 0.729] from raw  $3 \times 3$  tokens on the same section, a margin of +0.034 where the 1° pair gave +0.013. Same sign, roughly twice the size, still a small fraction of its own confidence interval, and — the part that governs how it should be quoted — *one seed*. We wrote above that no head number in this programme should be quoted from a single seed, and we decline to exempt this one because it happens to point the way we would prefer.

Honesty box: the head is the most expressive probe we run ( $\sim 25k$  parameters against  $\sim 220$  training months per fold), and although it is regularized hard, early-stopped on a train tail, and seed-averaged behind the same blocked folds as every other number, the head-vs-ridge gaps have not yet been given a paired significance test. The second-seed replication is no longer outstanding — it is reported above, and it cost us the pretraining margin. A capacity sweep of the head itself (23k  $\rightarrow$  325k  $\rightarrow$  2.0M parameters) shows *neither* side is parameter-limited: at 325k both columns drop (raw- $3 \times 3$  0.659  $\rightarrow$  0.590, embedding- $3 \times 3$  0.690  $\rightarrow$  0.611), as the label-count arithmetic predicts ( $\sim 220$  training months per fold support a head of tens of thousands of parameters, not hundreds), and the embedding margin persists

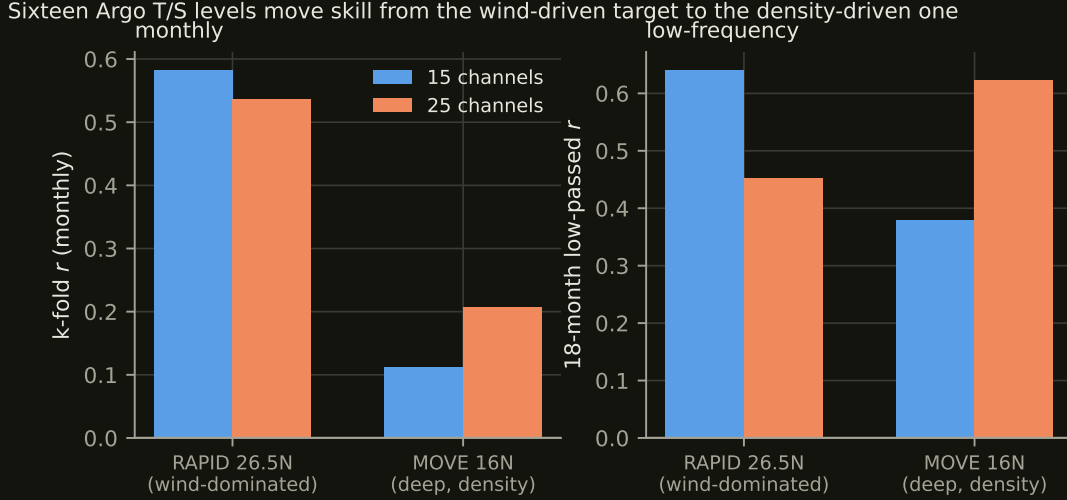


Figure 4: Sixteen Argo temperature and salinity levels to 1900 dbar *appeared* to cost monthly RAPID skill under pooled probes (left, first pair) — resolved by §7.3 as a read-out artifact, not a representation change — while genuinely roughly doubling skill at MOVE (left, second pair) and transforming the low-frequency read-out there (right). MOVE measures the *deep* western-boundary flow at 16°N — precisely the quantity the added levels describe.

at both sizes. The 2.0M cells are running as a formality.

## 7.4 First light from the quarter-degree tensor

The first codec trained on the quarter-degree tensor — a pilot-size (0.92M) control,  $C=39,84,405$  pixels, single seed — posts the highest pooled-ridge k-fold of the programme: RAPID  $r = 0.620$  [0.484, 0.741], RMSE 2.25 Sv, 18-month low-passed 0.747, 46% of the 2009–10 dip. Three readings, in decreasing confidence. First, the baseline discipline holds: the wind-only ridge measured *on this tensor* is 0.568, so the codec clears its own baseline by +0.052 where the deep-tensor pilots sat exactly on theirs — though the CIs overlap and this is one seed. Second, MOVE transfers again (0.235 [0.112, 0.340], low-passed 0.726) while wind-only at MOVE is *negative* (−0.376): at 16°N the embedding carries signal the wind actively gets wrong. Third, and to be read with care: 0.620 beats every 1° codec on the same truth and protocol, but across tensor generations that sentence means “this data recipe is better”, not “this codec is better” (§3.2).

The 41M-parameter codec trained *at* this tensor’s compute-optimal anchor has since finished, and its answer is a flat one: pooled k-fold RAPID  $r = 0.631$  [0.513, 0.732], RMSE 2.16 Sv, against the pilot’s 0.620 on the same tensor and protocol. Forty-four times the parameters bought +0.011, comfortably inside seed noise. Both clear the quarter-degree wind bar (0.568) and neither does so significantly. The honest reading is that skill on this tensor is *not* capacity-limited at the pooled read-out — which, combined with the flat 50k→1M step sweep of §10, closes both easy scaling axes at once: neither more parameters nor more steps is what stands between this programme and a better transport number. What did move: the unpooled head reads 0.662 [0.557, 0.745] from the same frozen codec, the 2009–10 dip capture rose to **51%** from the pilot’s 46%, and the 18-month low-passed correlation reached 0.820 — the last of which should be read against the  $\sim 9$  effective degrees of freedom of §5, not as a 0.82-with-240-months would be.

## 7.5 Where more data moved the skill

Under *pooled* probes, expanding the input appeared to cost RAPID skill: both deep-tensor codecs landed on the wind-only line (0.543 and 0.536 against 0.531) where the coarse tensor held +0.07. Section 7.3 resolves this: the skill had moved into cross-pixel structure, not out of the representation, and the unpooled read-out recovers it in full (0.617–0.690). The RAPID half of this subsection’s original claim is therefore retracted; what follows — the MOVE result — is the part that survives, and it was measured

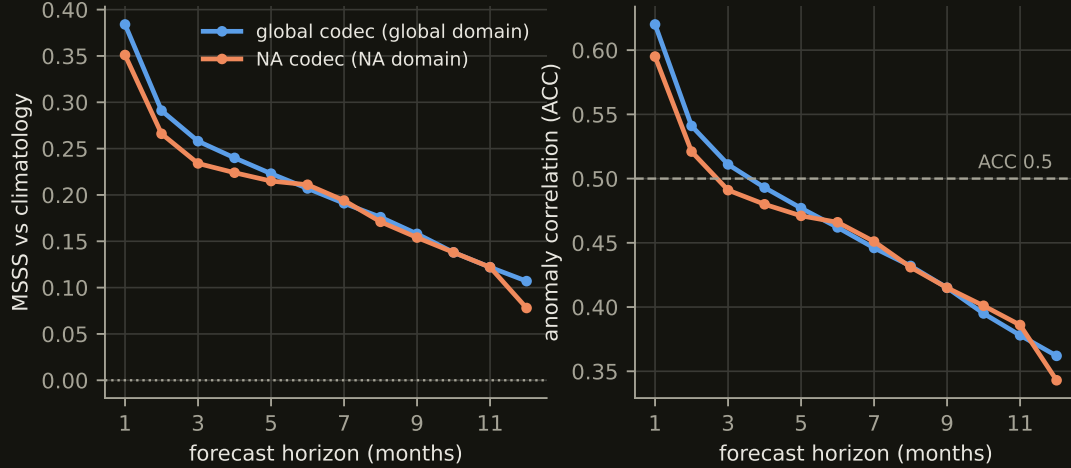


Figure 5: Autoregressive rollout skill versus forecast horizon for the global codec (blue, global domain) and the regional codec (orange, North-Atlantic domain; domains differ, so compare shapes more than levels). Left: MSSS against climatology — still positive at 12 months. Right: anomaly correlation, crossing the conventional 0.5 useful-skill line near four months.

pooled, so unpooling can only strengthen it.

At MOVE the same channels do the opposite. The 25-channel codec scores  $r = 0.206$  [0.044, 0.346] — the first multi-target result in this programme whose confidence interval excludes zero — with an 18-month low-passed correlation of 0.623, against 0.111 and 0.379 for the 15-channel codec. MOVE is a deep, density-driven transport; RAPID’s monthly variance is largely Ekman. The added channels are Argo  $T$  and  $S$  to 1900 dbar. The skill went where the physics says the information is.

The corrected reading is about read-outs, not bottlenecks: the embedding kept everything, and the probe determined what could be seen. For the generic-embedding hypothesis of §6 this is a *stronger* confirmation than the regional-versus-global comparison — one task-blind representation now serves the wind-dominated and the density-dominated target simultaneously, provided the read-out respects the structure of the question being asked of it.

One more asymmetry is worth recording: the 25-channel codec captures **59%** of the winter 2009–10 collapse out-of-fold, the best of any run in the programme, while its monthly correlation is the second worst. Event anatomy and average correlation are not the same capability, and a table that reports only the latter will hide the former.

## 7.6 How far ahead the ocean can be predicted

Fig. 5: feeding predictions back month after month, the anomaly field stays predictable beyond a year (MSSS +0.11 at  $h=12$ ; ACC 0.36). The decay is decorrelation, not amplitude damping (the forecast-to-observed amplitude ratio holds near 0.5 throughout). The AMOC read-out on rolled embeddings, by contrast, dies after about a season — transport forecasting at range remains open, consistent with its wind-driven monthly variance.

## 7.7 Case study: the winter 2009–10 collapse

# 8 The rolled forecast at scale

Everything above probes what the representation *contains*. This section — the work of 13–16 August, and the reason for v6 — is about what the temporal model can *do* with it: predict the ocean’s state forward, month over month, over the pixels that carry the overturning. One sentence of configuration so the reader carries the right picture: *everything in this section runs on the quarter-degree tensor* — all 39 channels of Table 1, including the sixteen Argo depth levels to 1900 dbar and sea-surface height — through the frozen 40.7M anchor codec of §7.4; the 14- and 24–25-channel tensors of the earlier sections

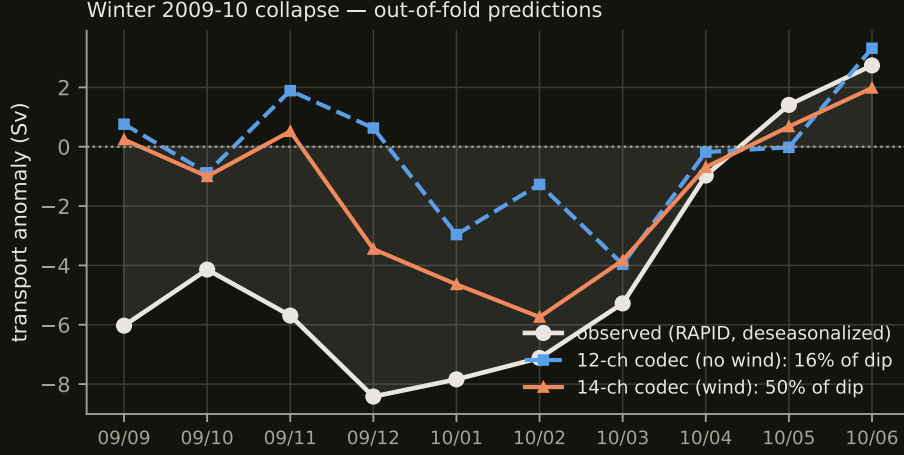


Figure 6: The sharpest event in the RAPID record, predicted out-of-fold. Without wind channels (blue) the model captures 16% of the dip amplitude; with them (orange), 50%. The 25-channel codec reaches 59% (not plotted; see Table 3). The November onset — triggered inside a month — stays invisible to monthly data.

are history here, and “scale” below always means the size of the *stage-2 temporal head*, never the codec or the channel count. The section’s one-sentence summary: input geometry looked decisive and was a symptom; **capacity was the variable**, and it has not stopped paying at 205M parameters.

## 8.1 Spatial coupling: the stencil

The stage-2 model of §4 predicts a pixel’s next embedding from that pixel’s own history — an ocean of independent columns. Physically that is wrong in a specific way: anomalies *adve*ct, and a month of Gulf Stream transport moves water hundreds of kilometres. The stencil gives the temporal model that information: alongside the centre pixel’s 24-month embedding sequence it reads the sequences of  $S$  neighbour pixels at fixed geographic offsets. Three geometry families were swept (rings at one radius; spirals, every point on its own bearing; and golden-angle *sunflower* layouts with a far-heavy  $\sqrt{\cdot}$  radial ramp at Fibonacci counts), with reach up to 4444 km. Reach beyond 4444 km was measured and *refused* before any training: on this window it deletes live input (26.1 live corridor slots at  $r_{\max}=4444$  km for a 55-point sunflower, 19.7 at 6666 km), because the extra reach lands on land or off-window.

Two scoreboards, defined in the glossary, are kept strictly apart — the **forecast ratio** (one step ahead, against persistence, reproducible to  $\pm 0.002$  across seeds) and the **corridor AUC** (twelve months rolled, feeding predictions back, scored where the AMOC lives). The programme’s founding observation (E-022, 13 August) was that they can *disagree*: a change that improves the one-step forecast can worsen what compounds under iteration. Every evaluation runs behind the **gate head** discipline — a frozen no-neighbour reference must reproduce its pinned corridor AUC of 0.643 exactly before any new number is accepted; it has now done so on every evaluation in this section.

## 8.2 An inversion at base scale, and the audit that cleared the instrument

At the then-standard stage-2 size ( $576 \times 8$ , 34M parameters), a factorial over stencil geometries produced a clean and disturbing result: across eleven arms the one-step and rolled scoreboards were *anti-correlated*. The widest, farthest-reaching stencils posted the best forecast ratios in the programme and rolled *worse than the no-neighbour baseline*; the best rolled arm was a minimal ring of 8 points at 222 km whose one-step forecast was mediocre. The full base-scale table is preserved in Appendix A.

Because an inversion of that shape could equally be a bug in the evaluator, it was audited before being believed (“investigate thoroughly and not hypothesize”). Six artifact classes were tested and eliminated on the archived evaluations alone, zero GPU spent: per-head scored-set differences (sample counts are identical to the pixel), broken rolled-input construction (at  $h=1$  the roll consumes observed truth and reproduces the training ordering exactly), baseline choice (ordering invariant under clima-

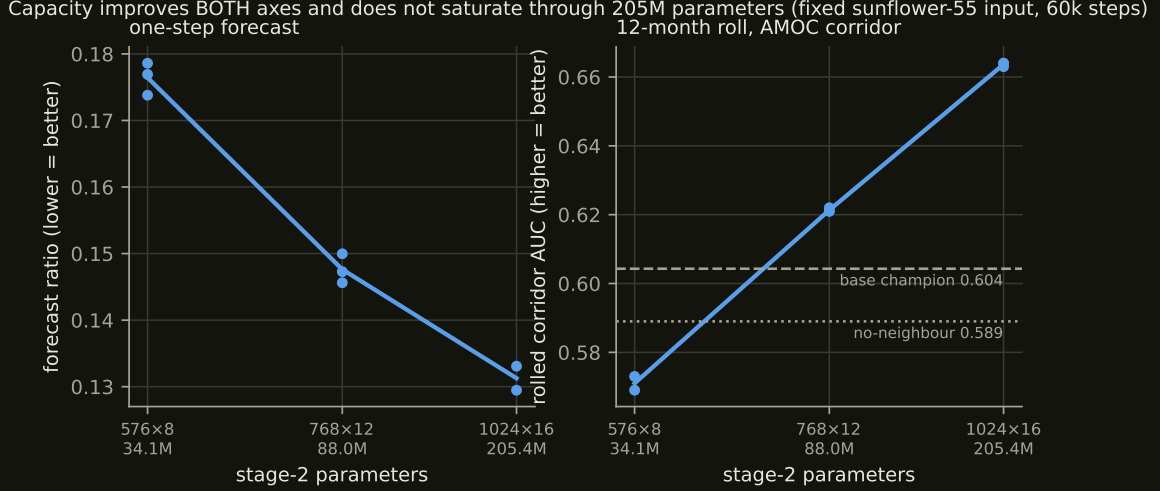


Figure 7: The capacity ladder at fixed input (sunflower-55, 60k steps), all seeds shown, parameters measured from the checkpoints. Left: the one-step forecast ratio. Right: the rolled corridor AUC, against the no-neighbour baseline (dotted) and the best base-scale geometry (dashed). The  $576 \times 8$  tier is *below* both reference lines — that is the inversion of §8.2 — and the same input geometry crosses both lines the moment the head is large enough. Neither panel shows saturation at 205M.

tology, persistence, and damped persistence), amplitude collapse (amplitude ratios do not track the inversion — the far-reach heads predict confident, full-amplitude, increasingly *wrong* patterns), horizon weighting (the top ranks are invariant), and boundary/dead-slot artifacts — eliminated with the sharpest measurement of the audit: the rolled penalty *rises* with the fraction of the stencil that is live ( $r = +0.176$ , monotone in quintiles), so pixels whose far stencils are fully over ocean pay most. An instrumented re-roll added channel structure: the penalty concentrates on the slow 300–700 m heat and salt reservoir — the AMOC-relevant fields — while the same wide stencils *keep* their advantage on fast wind-stress channels under iteration. The instrument was clean; the phenomenon lived in the model-under-iteration.

### 8.3 Capacity breaks the inversion

The resolving experiment was a  $2 \times 2$  — head size ( $576 \times 8$  vs  $768 \times 12$ )  $\times$  input width (34 vs 55 sunflower points), three seeds per cell, everything else frozen. On the forecast ratio the capacity main effect is  $-0.029$  paired at every seed ( $\sim 30\times$  the three-seed noise bar); the width effect is  $-0.002$ , same sign at every seed; and there is *no interaction* — width helps by the same small amount at both sizes. Capacity is worth three times the entire geometry axis of Appendix A.

The decisive question was whether that gain *rolls* — a one-step gain that evaporated under iteration would have been exposure bias wearing a scaling law’s clothes. It rolls: paired by seed, the  $768 \times 12$  tier gains **+0.050** corridor AUC over  $576 \times 8$  on the same 55-point sunflower ( $+0.052/+0.048$ ), and the same geometry that rolled at  $0.569\text{--}0.573$  (below the no-neighbour  $0.589$ ) rolls at  $0.621\text{--}0.622$  — past the no-neighbour baseline *and* past the base-scale champion’s  $0.604$ . The E-026 inversion was a capacity-starvation artifact: a 34M head fed  $55 \times 64$ -dimensional inputs learns a one-step mapping whose errors compound under iteration; an 88M head learns one whose errors do not. The channel fingerprint of the audit (slow subsurface pays, wind survives) is what that starved mapping looks like.

One tier up ( $1024 \times 16$ , 205.4M, same input, same steps), both scoreboards improve again: forecast ratio  $0.1295\text{--}0.1331$  (the project’s best), corridor AUC **0.664**/ $0.663/0.664$  —  $+0.042$  over the 88M tier,  $+0.075$  over the no-neighbour baseline, with a seed spread of  $0.001$  against the forecast axis’s  $0.004$ : at this capacity the rolled skill is essentially deterministic given the data. Fig. 7 shows the ladder; Table 4 is the per-seed record. No saturation is visible on either axis through 205M parameters.

| head                                                                       | tier             | s0           | s1           | s2    | mean          |
|----------------------------------------------------------------------------|------------------|--------------|--------------|-------|---------------|
| no neighbours (gate)                                                       | $576 \times 8$   | 0.589        | —            | —     | 0.5890        |
| base55 (sunflower-55)                                                      | $576 \times 8$   | 0.569        | 0.573        | —     | 0.5710        |
| ring-8@222 (base champion)                                                 | $576 \times 8$   | 0.608        | 0.599        | 0.606 | 0.6043        |
| big55 (sunflower-55)                                                       | $768 \times 12$  | 0.621        | 0.621        | 0.622 | 0.6213        |
| big34 (sunflower-34)                                                       | $768 \times 12$  | 0.628        | 0.618        | 0.625 | 0.6237        |
| x155 (sunflower-55)                                                        | $1024 \times 16$ | 0.664        | 0.663        | 0.664 | 0.6637        |
| <i>not a capacity change — same 88M tier, different input or objective</i> |                  |              |              |       |               |
| ring-8@222 (near-field)                                                    | $768 \times 12$  | 0.646        | 0.661        | 0.661 | 0.6560        |
| znoise, $\sigma=0.7$ (sun-55)                                              | $768 \times 12$  | <b>0.683</b> | <b>0.674</b> | —     | <b>0.6785</b> |

Table 4: **Rolled corridor AUC, per seed** — the reference table for the metric defined in §2. Every row is read from an evaluation whose gate head reproduced its pinned 0.643 exactly; the full per-horizon and per-channel outputs are archived per evaluation run (`ml-metrics` branch). The upper block is the capacity ladder at fixed input. **The lower block is the result that reorders this section:** neither row is a capacity change, and both beat the tier they belong to — input noise by +0.057 and near-field geometry by +0.035 over the same 88M model, with the *noised 88M* model beating the *clean 205M* one outright. The base-scale geometry table behind the ring-8@222 row is Appendix A; the width and step-budget rows await the evaluations still in flight.

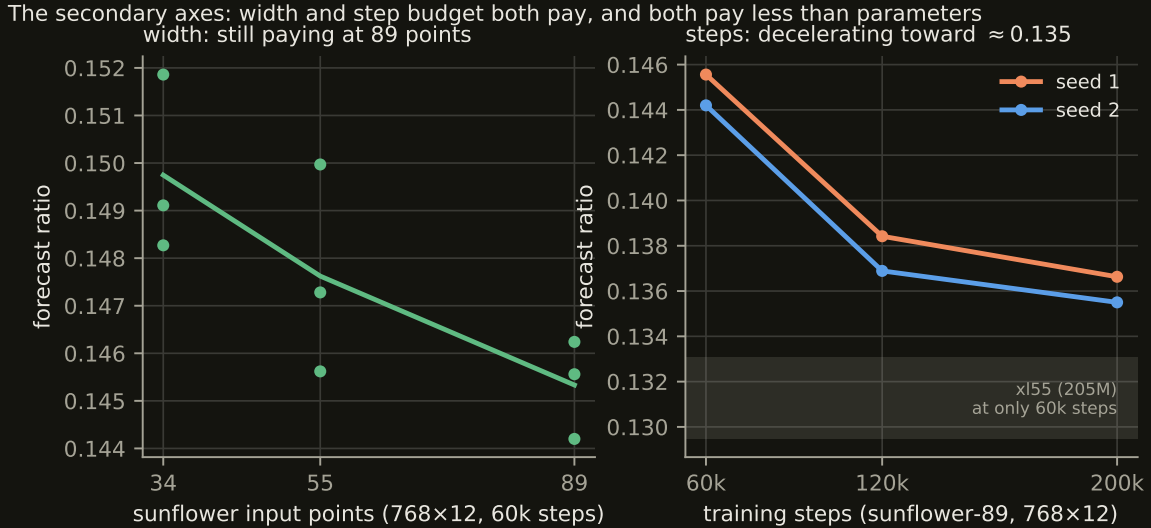


Figure 8: Left: input width at the 88M tier — the forecast ratio improves by  $\approx 0.002$  per Fibonacci rung and has not saturated at 89 points. Right: step budget at 89 points — the first 60k-step extension buys  $-0.0072$  paired, the second (80k steps) only  $-0.0016$ ; the trajectory is bending toward  $\approx 0.135$ , and the 205M tier at 60k steps (grey band) remains out of reach of the 88M tier at any step budget measured.

## 8.4 The other axes: width and step budget

With capacity established as the first-order variable, the remaining axes were measured against it (Fig. 8). **Width:** at the 88M tier, sunflower inputs of 34, 55, and 89 points give forecast ratios of  $0.1498 \rightarrow 0.1476 \rightarrow 0.1453$  (means of three seeds; paired per rung  $\approx -0.002$ , same sign at every seed) — small, consistent, unsaturated. **Step budget:** extending the 89-point arms from 60k to 120k to 200k steps under a horizon-free exponential-decay schedule traces  $0.1449 \rightarrow 0.1377 \rightarrow 0.1361$  (two-seed means), i.e. a per-step value that falls  $\sim 6\times$  between the first extension and the second. The comparison that matters: the 88M/89-point model at its *full* 200k budget (0.1355–0.1366) still does not reach the 205M model at 60k (0.1295–0.1331). At this scale, *parameters beat steps*, which is the same verdict the corridor gave. Runs combining the widest inputs with the largest head (89 and 144 points at  $1024 \times 16$ , 200k steps) are in flight at the time of writing, as are the rolled corridor evaluations of the width and step-budget arms.

## 8.5 Exposure bias: two mechanisms under test

The audit’s finding — errors that compound under iteration — suggests the classic exposure-bias remedies, and two are under test. *Input-noise training* adds Gaussian noise to the input embeddings at the model’s own measured one-step error scale ( $\sigma = 0.7$   $z$ -units), so training-time context statistically resembles roll-time context; as predicted it *costs* one-step forecast (0.1548/ 0.1539 vs. the clean tier’s 0.1476), and whether it buys the roll is being evaluated now. *One-hop wide unroll* exploits a dependency-cone observation: to train through one step of self-generated context at stencil width  $S$ , one does not need to roll the whole ocean — only the  $S$  input pixels of the next step, each of which is exactly a depth-1 prediction from its own observed window. This is implemented (detached depth-1 forwards over the slot pixels, an assembled  $t+1$  window, a differentiable second step against  $Z_{t+2}$  at half weight) and verified by alignment tests; its first two arms were stopped mid-run by an operator decision after the measured step cost came in at  $8.5\times$  the clean model (gather-dominated), and the checkpoint state is preserved for a decision on resumption. Neither mechanism has yet produced a rolled verdict; both are falsifiable against the clean 88M tier’s corridor AUC of 0.6213.

## 8.6 The dependency cone, and what every rolled number assumes

A stencil head predicts a pixel from its  $S$  neighbours, so a two-step prediction depends on the neighbours’ neighbours, and the set of pixels an  $h$ -step prediction depends on — its *cone* — grows by the stencil’s reach every step. At 4444 km that is 4444 $h$  km against Earth’s 40,075 km circumference, so the cone wraps the planet within a few steps while our window is a basin. The evaluator already rolls every one of the 84,405 window ocean pixels, which is the most it can do; the question is whether the window is large enough.

Measured rather than argued (pure geometry over the real neighbour table, seeded from the RAPID section): for the 144-point sunflower the cone covers 26.9% of the window at  $h=1$ , 99.5% at  $h=2$  and **100% at  $h=3$** , after which  **$\sim 50\%$  of every step’s requested neighbours resolve to land or off-window**. The near-field ring-8@222 reaches only 30.7% of the window by  $h=12$  with 4.8% of its demand unmet — so it is *reach*, not slot count, that escapes.

This is not an error: dead slots are encoded as zero in training exactly as in evaluation, so no number here is invalidated. But it means every rolled result in this report carries a boundary condition we had not stated — **the world outside the window is held at its climatological mean** — and that the fraction of a model’s input owned by that assumption grows with its reach. It is the most plausible single explanation for §8.7’s two surprises: a wide-and-far model rolls on a state that is half boundary assumption, a near-field one barely depends on the boundary, and input noise makes a model robust to exactly the kind of corrupted context the assumption produces. It also promotes a global tensor from a data-volume argument to a correctness requirement for long rolls at wide reach.

## 8.7 Standings

| arm                    | tier             | params | input pts | forecast ratio       | corridor AUC  |
|------------------------|------------------|--------|-----------|----------------------|---------------|
| xl55 (60k)             | $1024 \times 16$ | 205.4M | 55        | <b>0.1295–0.1331</b> | <b>0.6637</b> |
| sun89 (200k)           | $768 \times 12$  | 89.7M  | 89        | 0.1355–0.1366        | in flight     |
| sun89 (120k)           | $768 \times 12$  | 89.7M  | 89        | 0.1369–0.1384        | —             |
| sun89 (60k)            | $768 \times 12$  | 89.7M  | 89        | 0.1442–0.1462        | in flight     |
| big55 (60k)            | $768 \times 12$  | 88.0M  | 55        | 0.1456–0.1500        | 0.6213        |
| big34 (60k)            | $768 \times 12$  | 86.9M  | 34        | 0.1483–0.1519        | 0.6237        |
| big-ring222 (60k)      | $768 \times 12$  | —      | 8         | 0.1509–0.1557        | <b>0.6560</b> |
| <b>znoise-55 (60k)</b> | $768 \times 12$  | 88.0M  | 55        | 0.1539–0.1548        | <b>0.6785</b> |
| base55 (60k)           | $576 \times 8$   | 34.1M  | 55        | 0.1738–0.1786        | 0.5710        |
| ring-8@222 (base)      | $576 \times 8$   | —      | 8         | 0.1848               | 0.6043        |
| no neighbours (gate)   | $576 \times 8$   | —      | 0         | —                    | 0.5890        |

Table 5: **Stage-2 standings, both scoreboards**, 16 August 2026. Forecast ratio ranges are seed min–max; corridor AUC is the seed mean, every value behind a passed gate. “In flight” marks evaluations running at the time of writing. The two scoreboards now *agree* at adequate capacity — the anti-correlation of Appendix A is confined to the 34M tier.

Table 5 is the current state of the programme, and the rolled column has just overturned the reading this section was drafted with. Capacity dominates the *forecast* axis and pays handsomely on the roll — both of those stand. What is no longer true is that capacity dominates *everything*: at a fixed 88M, adding Gaussian noise to the model’s own inputs is worth +0.057 corridor AUC and beats the 205M model, while a near-field 8-point stencil is worth +0.035 and also beats every wide-input arm of its own tier. The two cheapest interventions in the programme outrank its most expensive one.

The honest summary is therefore a rebalanced one. On one-step forecast, scale is the whole story and the axes are ordered parameters  $\gg$  width  $>$  steps. On the twelve-month roll, *robustness to the model’s own error* is at least as valuable as parameters, and the ranking of input geometries does not follow the forecast ranking at all — the two scoreboards disagree again (§8.2) at a capacity where starvation cannot explain it. §8.6 gives what we think is the underlying reason. The next experiments are the compositions nobody has run: noise  $\times$  205M, noise  $\times$  near-field, and a sweep of  $\sigma$ , which was set from the model’s measured one-step error and never tuned.

## 9 The static-channel lesson

An ablation found the two climatology input channels contribute *exactly nothing* to any metric: a static map is a function of position, the encoder receives position, and so the channel carries no information the model does not already have. The general rule — a channel is worth what position and season cannot explain of it — now gates every data-acquisition decision. Climatological knowledge is still essential, but it enters as preprocessing arithmetic (the anomaly transform), never as a feature the network must carry through its bottleneck.

## 10 Limitations and open questions

(1) The wind-only baseline bounds honest claims: monthly AMOC skill is substantially Ekman, though the unpooled head now clears the baseline by +0.10–0.16 where the pooled probes had shrunk the margin to noise. (2) The head result is new and strong, and strength is when scrutiny matters most. The second codec seed has now been run and it removed the pretraining margin (§7.3); what remains outstanding is a paired significance test, and the fact that the head has been run only on RAPID — the MOVE and low-frequency versions may move too. The head is also demonstrably the noisiest probe we own (seed spread 0.036 against the ridge’s 0.012), so single-seed head numbers should not be quoted anywhere, including by us. (3)  $d_z=64$  was chosen on a 12-channel tensor and carried unchanged into 24–25 channels; the sweep has not been repeated at the new width. (4) Absolute k-fold values are mildly optimistic — the codec saw non-holdout months’ *fields* during self-supervised pretraining (transports were never seen by anything); codec-to-codec comparisons are unaffected. (5) The rollout’s horizon is measured only to the holdout-year boundary; the crossover to no skill lies beyond 12 months and is not yet located. (6) Transport forecasting at range is unsolved. (7) *A methodological casualty worth recording*: the stage-2 results for two runs were withdrawn after we found their embedding cache had been populated by a different codec — the shape check  $(T, P, d_z)$  matched, so training proceeded happily on the wrong representation, and the signature was healthy embedding-space skill beside catastrophic decoded skill. The cache key now carries a hash of the codec’s weights, so a stale cache is a miss rather than a lie. We report the affected numbers as withdrawn rather than regenerating them, because both codecs belong to the superseded coarse tensor.

For the rolled-forecast programme of §8, four things are open at the time of writing, all mechanical rather than conceptual: the rolled corridor evaluations of the width, step-budget, and exposure-bias arms are in flight (their forecast-axis numbers are final); the widest-input  $\times$  largest-head runs (89 and 144 points at 205M, 200k steps) land within hours; the one-hop unroll arms await a resume decision; and the capacity ladder has no measured top — every tier so far improved both scoreboards, and the next tier is the programme’s standing next experiment. The corridor AUC itself carries one structural caveat: it scores the model’s own input fields rolled forward, not transport — the bridge from rolled field skill to transport skill at range is the same open problem as (6) below.

The training-time ceiling has now been located directly: a 1,000,000-step run at pilot size (LR decayed to a constant 10% floor by 50k steps, probes every 50k) completed in 2.6 h on one RTX 4090 and is *flat from the first probe to the last* — linear  $r$  oscillates in  $[0.43, 0.48]$ , channel-space skill vs. persistence is pinned at 32%, and the final checkpoint’s k-fold (0.571) sits within seed noise of the 40k-

step run. Twenty-five times the compute of the pilot regime buys nothing at 0.92M parameters: the pilot is capacity-limited, which is what the Chinchilla arithmetic predicted ( $\sim 13$ M parameters for this tensor) and what the 10M-parameter codecs now training are for. Next levers, in expected-value order: paired significance for the head results; the head probe at MOVE and at low frequency; re-running the  $d_z$  sweep at  $C=24$ ; then the approved data ladder — (a) extending the time axis to 1982, where monthly SST and wind exist pre-GLORYS and the Florida cable’s 1982–92 decade becomes truth, (b)  $0.25^\circ$  resolution for the GLORYS-derived channels ( $16\times$  the pixels), (c) doubling the Argo pressure levels — with the Chinchilla arithmetic redone and the codec re-sized at each rung; and label-efficiency curves over the 792 non-RAPID truth months.

## A Base-scale results superseded by capacity

Everything in this appendix was measured at the  $576 \times 8$  (34M) stage-2 tier, and §8.3 showed that tier to be capacity-starved for wide inputs: the geometry ranking below is a property of a too-small head, not of the ocean. It is preserved because the numbers are real, the discipline that produced them (pre-registered hypotheses, paired seeds, gate reproduction) is the same as the main text’s, and because a ranking this clean that *dissolved* under one scaling step is itself a finding worth keeping in view.

### A.1 The geometry factorial at 34M

Eleven stencil arrangements, three seeds each where completed, rolled corridor AUC (every evaluation gate-passed):

| stencil                | pts | reach km | AUC mean      | vs champion |
|------------------------|-----|----------|---------------|-------------|
| <b>ring-8 @222</b>     | 8   | 222      | <b>0.6043</b> | 0           |
| spiral-8               | 8   | 890      | 0.5883        | −0.016      |
| no neighbours          | 0   | 0        | 0.5890        | −0.015      |
| three rings narrow     | 12  | 1000     | 0.589         | −0.015      |
| spiral-13              | 13  | 1000     | 0.5863        | −0.018      |
| two rings 8+8          | 16  | 555      | 0.5857        | −0.019      |
| three rings wide       | 24  | 1000     | 0.5813        | −0.023      |
| spiral-24              | 24  | 4444     | 0.578         | −0.026      |
| elliptic-24            | 24  | 4444     | 0.568         | −0.036      |
| spiral-34              | 34  | 4444     | 0.566         | −0.038      |
| ring-16 @222 ( $n=1$ ) | 16  | 222      | 0.564         | −0.040      |
| sunflower-34           | 34  | 4444     | 0.553         | −0.051      |

The ranking is monotone in slot count within the spiral family ( $8 \rightarrow 13 \rightarrow 24 \rightarrow 34$  gives  $0.588 \rightarrow 0.586 \rightarrow 0.578 \rightarrow 0.566$ ), and two isolations pin the axes: doubling points at fixed 222 km radius costs  $-0.044$ ; widening reach at fixed 8 points costs  $-0.016$ . Meanwhile the *forecast* ordering of the same arms ran the other way — the sunflower and deep-spiral arms were the best one-step models in the programme at the time. That anti-correlation, audited clean in §8.2, is what §8.3 resolved: at 88M and 205M the wide sunflower geometry wins *both* scoreboards, and the production-geometry question was re-opened at scale rather than settled by this table. The base-scale champion’s ring-8@222 configuration is itself being re-evaluated at the 88M tier at the time of writing.

### A.2 Early stage-2 capacity sweep and codec training curves

The original stage-2 capacity sweep (Fig. 9, left) was run on the coarse tensor at 0.11–4M parameters and read, at the time, as “capacity buys dynamics.” It deserved more weight than it got: an early experiment at one slot and small  $K$  (“E-010”) found stage-2 skill insensitive to capacity and closed the axis; that negative was real *in its regime* (a 1-slot input barely exercises a large head) and wrong outside it, and re-opening the axis at  $K=24 \times 55$  slots is what produced §8.3. A settled negative is settled only in the regime that settled it.

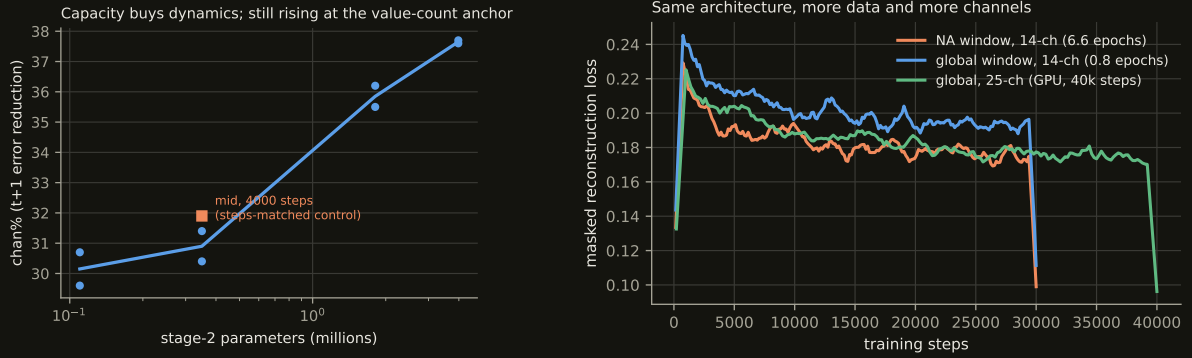


Figure 9: Left: the original stage-2 capacity sweep (0.11–4M parameters, 12-channel coarse tensor, no stencil) — the first sign, in retrospect, that stage-2 capacity was the binding axis; it was still rising at its top end. Right: codec training curves across generations.

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